

Exercises from Ch. 1: Groups, Lang

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Exercises

Exercise 0.1. (Exercise 1) Show that every group of order ≤ 5 is abelian.

Proof. First, we will take care of the cases where $|G|$ is prime (i.e. 2, 3, 5). We prove this by showing that the groups of these orders are cyclic and thus abelian (cyclic $\Rightarrow$ abelian). Suppose that $|G| = p \leq 5$, a prime. Then, by Lagrange's Theorem, for all $x \in G$, $|x| \mid |G|$. Hence, $|x| \mid p$. Therefore, $|x| = 1$ or $|x| = p$. If $|x| = 1$, then $x = e$ and we get $|G| = 1$ which is not a prime, but is also trivially abelian. If $|x| = p$, then let $G = \langle x \rangle$. Thus, $G = \{e = x^{|G|}, x, x \cdot x = x^2, x \cdot x \cdot x = x^3, \dots, x^{|G|-1}\}$. Clearly, G is abelian since $x \cdot x^j = x^j \cdot x$ for all $x \in G$ and $1 \leq j \leq |G|$. Thus, if $|G| \leq 5$ is prime, then G is abelian. Next suppose $|G| = 4 = 2^2$. We prove a general case that all groups of order p^2 for a prime p are abelian and thus it would follow that G is abelian if $|G| = 4$. Suppose $|G| = p^2$ for a prime p . Hence, G is a p -group and hence $Z(G)$ is not trivial. Recall that the class equation is defined as

$$|G| = |Z(G)| + \sum_{x \in C} (G : G_x)$$

where C is a set of representatives for the distinct conjugacy classes and G_x is the isotropy group of x in G defined by $G_x = \{y \in G : x \cdot y = y\}$. But here we can consider G acting on itself by conjugation and thus $G_x = \{y \in G : xyx^{-1} = x\} = C_G(x) = \{g \in G : gx = xg\}$. Each term in $\sum_{x \in C} (G : G_x) > 1$ and divides $|G| = p^2$. Hence, $\sum_{x \in C} (G : G_x) \equiv 0 \pmod{p}$. Therefore, $|G| = |Z(G)| + kp$ ($k \in \mathbb{Z}^+$, and

$kp = \sum_{x \in C} (G : G_x)$). Then, $|Z(G)| = |G| - kp \Rightarrow |Z(G)| \equiv 0 \pmod{p}$. Thus, $p \mid |Z(G)| \Rightarrow |Z(G)| = p$ or p^2 (since $|Z(G)| \leq |G|$).

Case 1: Suppose $|Z(G)| = p^2$. Then $|Z(G)| = |G| \Rightarrow Z(G) = G$. Thus, G is abelian.

Case 2: Suppose $|Z(G)| = p$. Consider the quotient group $G/Z(G)$ (since $Z(G) \triangleleft G$). Thus,

$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{p^2}{p} = p$. Hence, $G/Z(G)$ is cyclic by above and thus $G/Z(G)$ is abelian. Now we will show that the fact that $G/Z(G)$ is abelian implies that G is abelian.

Since $G/Z(G)$ is cyclic, $G/Z(G) = \langle gZ(G) \rangle$. Thus, $xZ(G) = (gZ(G))^k = g^k Z(G)$ for some $k \in \mathbb{Z}$. By coset equality, $xZ(G) = g^k Z(G) \iff x(g^k)^{-1} \in Z(G) \iff x = g^k z_1$ for some $z_1 \in Z(G)$. Hence, every element of $x \in G$ can be written as $x = g^k z_1$ for some $k \in \mathbb{Z}$ and $z_1 \in Z(G)$. Similarly, let $y \in G$ and thus $y = g^\ell z_2$ for some $\ell \in \mathbb{Z}$ and $z_2 \in Z(G)$. We want to show that $xy = yx$. Then,

$$xy = g^k z_1 g^\ell z_2 = g^{k+\ell} z_1 z_2 = g^{\ell+k} z_2 z_1 = g^\ell z_2 g^k z_1 = yx$$

Therefore, G is abelian and note that we were able to commute elements since $z_1, z_2 \in Z(G)$. Thus, we have shown that if $G/Z(G)$ is cyclic, then G is abelian. We are done. □

Exercise 0.2. (Exercise 2)

Show that there are two non-isomorphic groups of order 4, namely the cyclic one, and the product of two cyclic groups of order 2.

Proof. Let G be a group of order 4. Let $g \in G$ such that $g \neq e$. By Lagrange's Theorem, $|g| \mid |G|$. Thus, $|g| = 2$ or 4. If $|g| = 4$, then $G = \langle g \rangle \cong \mathbb{Z}/4\mathbb{Z}$. If $|g| = 2$, then $G = \langle g \rangle \times \langle g \rangle \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. These are non-isomorphic since one is cyclic and one is not. $\square$

Exercise 0.3. (Exercise 3)

Let G be a group. A **commutator** in G is an element of the form $aba^{-1}b^{-1}$ with $a, b \in G$. Let G^c be the subgroup generated by the commutators. Then G^c is called the **commutator subgroup**. Show that G^c is normal. Show that any homomorphism of G into an abelian group factors through G/G^c .

Proof. By definition, $G^c = \{[a_1, b_1]^{c_1} \cdots [a_n, b_n]^{c_n} : n \in \mathbb{N}, a_i, b_i \in G, c_i = \pm 1\}$. Thus, by construction, $G^c \leq G$. We show that this subgroup is normal. First, observe that for a single commutator,

$$\begin{aligned} g[a, b]g^{-1} &= g(aba^{-1}b^{-1})g^{-1} \\ &= ga(g^{-1}g)b(g^{-1}g)a^{-1}(g^{-1}g)b^{-1}g^{-1} \\ &= (gag^{-1})(gbg^{-1})(ga^{-1}g^{-1})(gb^{-1}g^{-1}) \\ &= [gag^{-1}, gbg^{-1}] \in G^c \end{aligned}$$

Conjugation is a homomorphism (distributes over products), and thus, since an element of G^c is a finite product of commutators, we have

$$\begin{aligned} g([a_1, b_1]^{c_1} [a_2, b_2]^{c_2} \cdots [a_n, b_n]^{c_n})g^{-1} &= g((a_1 b_1 a_1^{-1} b_1^{-1}) \cdots (a_n b_n a_n^{-1} b_n^{-1}))g^{-1} \\ &= g(a_1 b_1 a_1^{-1} b_1^{-1})g^{-1} g(a_2 b_2 a_2^{-1} b_2^{-1})g^{-1} \cdots g(a_n b_n a_n^{-1} b_n^{-1})g^{-1} \\ &= (g[a_1, b_1]g^{-1}) (g[a_2, b_2]g^{-1}) \cdots (g[a_n, b_n]g^{-1}) \end{aligned}$$

and our previous remarks, for all $1 \leq i \leq n$, $g[a_i, b_i]g^{-1} = [ga_i g^{-1}, gb_i g^{-1}]$. Then we get

$$g([a_1, b_1] \cdots [a_n, b_n])g^{-1} = [ga_1 g^{-1}, gb_1 g^{-1}] \cdots [ga_n g^{-1}, gb_n g^{-1}] \in G^c$$

Note that since conjugation preserves products and inverses, we do not necessarily need to address the exponents $c_i = \pm 1$. Thus, $gG^c g^{-1} \subseteq G^c$ for all $g \in G$. Hence $G^c \triangleleft G$.

Let $f: G \rightarrow A$ be a homomorphism of G into an abelian group A . We want to show that the following diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{f} & A \\ & \searrow \pi & \nearrow \bar{f} \\ & G/G^c & \end{array}$$

Here π is the canonical homomorphism (surjective). Observe that since f is a homomorphism and A is abelian,

$$\begin{aligned} f([a, b]) &= f(aba^{-1}b^{-1}) = f(a)f(b)f(a)^{-1}f(b)^{-1} \\ &= f(a)f(a)^{-1}f(b)f(b)^{-1} \\ &= e. \end{aligned}$$

Since every $x \in G^c$ is a finite product of commutators, by the fact that f is a homomorphism and by the above computation,

$$f(x) = e \quad \forall x \in G^c$$

Define

$$\bar{f}: G/G^c \rightarrow A$$

by $gG^c \mapsto f(g)$. First, we show that $\bar{f}$ is well-defined. Suppose $gG^c = hG^c$. Thus, by coset equality, $gh^{-1} \in G^c$. Then

$$\begin{aligned} f(gh^{-1}) &= f(g)f(h)^{-1} = e \\ &\Rightarrow f(g) = f(h) \end{aligned}$$

Therefore, $\bar{f}$ is well-defined. Now, let $gG^c, hG^c \in G/G^c$. Then

$$\begin{aligned} \bar{f}((gG^c)(hG^c)) &= \bar{f}(ghG^c) \\ &= f(gh) \\ &= f(g)f(h) \\ &= \bar{f}(gG^c)\bar{f}(hG^c) \end{aligned}$$

Thus, $\bar{f}$ is a homomorphism. Hence, for all $g \in G$,

$$(\bar{f} \circ \pi)(g) = \bar{f}(\pi(g)) = \bar{f}(gG^c) = f(g)$$

Thus, $\bar{f} \circ \pi = f$. Therefore, every homomorphism of G into an abelian group factors through G/G^c . $\square$

Exercise 0.4. (Exercise 4)

Let H, K be subgroups of a finite group G with $K \subset N_H$. Show that

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|}.$$

Proof. Since $K \subseteq N_H$, we have $HK \leq G$. Define

$$f: H \times K \rightarrow HK$$

by $(h, k) \mapsto hk$. Let $x \in HK$. Then $x = hk$ for some $h \in H$ and $k \in K$. Hence, $f((h, k)) = hk = x$.

Thus, f is surjective. Let $(h, k) \in \ker f$. Then

$$\begin{aligned} hk &= e \\ k &= h^{-1} \in K \end{aligned}$$

Thus, since $K \leq G$ and $h^{-1} \in K$, we must also have $h \in K$. Therefore

$$\ker f = \{(h, h^{-1}) \in H \times K : h \in H \cap K\} \triangleleft H \times K$$

We can define a bijection between $H \cap K$ and $\ker f$ by $h \mapsto (h, h^{-1})$. Thus, $|\ker f| = |H \cap K|$. By the First Isomorphism Theorem,

$$\begin{aligned} H \times K / \ker f &\cong f(H \times K) \\ H \times K / H \cap K &\cong HK \end{aligned}$$

Thus,

$$\begin{aligned} |H \times K / H \cap K| &= |HK| \\ \frac{|H \times K|}{|H \cap K|} &= |HK| \\ \frac{|H| \cdot |K|}{|H \cap K|} &= |HK| \end{aligned}$$

$\square$

Exercise 0.5. (Exercise 6)

Prove that the group of inner automorphisms of a group G is normal in $\text{Aut}(G)$.

Proof. Let $x \in G$. An inner automorphism is defined by $c_x(y) = xyx^{-1}$ for all $y \in G$. Something to note, perhaps irrelevant, is $f: G \rightarrow \text{Aut}(G)$ by $x \mapsto c_x$ is a homomorphism onto $\text{Inn}(G) \subseteq \text{Aut}(G)$ with kernel $Z(G)$, so by the First Isomorphism Theorem,

$$G/Z(G) \cong \text{Inn}(G) \subseteq \text{Aut}(G).$$

Let $\varphi \in \text{Aut}(G)$ and $c_x \in \text{Inn}(G)$ for some $x \in G$. Let $y \in G$. Then

$$\begin{aligned} (\varphi c_x \varphi^{-1})(y) &= \varphi(c_x(\varphi^{-1}(y))), \quad \varphi^{-1}(y) \in G \quad \because \varphi \in \text{Aut}(G) \\ &= \varphi(x\varphi^{-1}(y)x^{-1}) \\ &= \varphi(x)\varphi(\varphi^{-1}(y))\varphi(x^{-1}) \\ &= \varphi(x)y\varphi(x)^{-1} \\ &= c_{\varphi(x)}(y) \in \text{Inn}(G), \quad \because \varphi(x) \in G. \end{aligned}$$

Hence, $\text{Inn}(G) \triangleleft \text{Aut}(G)$. □

Exercise 0.6. (Exercise 7)

Let G be a group such that $\text{Aut}(G)$ is cyclic. Prove that G is abelian.

Proof. Suppose $\text{Aut}(G)$ is cyclic. Hence, $\text{Inn}(G)$ is cyclic since $\text{Inn}(G) \subseteq \text{Aut}(G)$. Consider the map $\psi: G \rightarrow \text{Inn}(G)$ given by $g \mapsto \varphi_g$ for all $g \in G$ where $\varphi_g: G \rightarrow G$ by $x \mapsto gxg^{-1}$ for all $x \in G$. Notice that $\text{Ker } \psi = Z(G)$, since if $\varphi_g(x) = gxg^{-1} = x$, then $gx = xg \Rightarrow g \in Z(G)$. Thus, by the First Isomorphism Theorem, $G/\text{Ker } \psi \cong \text{Inn}(G) \Rightarrow G/Z(G) \cong \text{Inn}(G)$. Since we have this isomorphism, $G/Z(G)$ and $\text{Inn}(G)$ have the same group structure, i.e. $G/Z(G)$ is cyclic and thus $G/Z(G)$ is abelian which implies from **Exercise 1** that G is abelian. □

Exercise 0.7. (Exercise 9)

- a) Let G be a group and H a subgroup of finite index. Show that there exists a normal subgroup N of G contained in H and also of finite index. [Hint: If $(G:H) = n$, find a homomorphism of G into S_n whose kernel is contained in H .]
- b) Let G be a group and let H_1, H_2 be subgroups of finite index. Prove that $H_1 \cap H_2$ has finite index.

Proof. a) Let G be a group and $H \leq G$ such that $(G:H) = n$. Consider the left cosets $G/H = \{g_1H, \dots, g_nH\}$. Define an action on these cosets by

$$\varphi_g: G/H \rightarrow G/H$$

by $\varphi_g(xH) = gxH$. This is well-defined for if $xH = yH$, then $x - y \in H$ and hence $gx - gy \in H \Rightarrow gxH = gyH \Rightarrow \varphi_g(xH) = \varphi_g(yH)$. The homomorphism φ_g can be shown to be bijective as well. Thus, $\varphi \in \text{Sym}(G/H) \cong S_n$. Define

$$\psi: G \rightarrow S_n \cong \text{Sym}(G/H)$$

by $\psi(g) = \varphi_g$. This is a homomorphism since for $g_1, g_2 \in G$,

$$\begin{aligned} \psi(g_1g_2) &= \varphi_{g_1g_2}(xH) \quad \text{for any } xH \in G/H \\ &= g_1g_2xH \\ &= g_1\varphi_{g_2}(xH) \\ &= \varphi_{g_1}(\varphi_{g_2}(xH)) \\ &= \varphi_{g_1} \circ \varphi_{g_2} \end{aligned}$$

The kernel, $\text{ker } \psi$, is defined to be $g \in G$ such that $\psi(g) = \text{id}$. Hence, if $\psi(g) = \text{id}$ (i.e. $g \in \text{ker } \psi$), we get

$$\begin{aligned}
\psi(g) &= id \\
\Rightarrow \varphi_g(xH) &= xH \quad \forall xH \in G/H \\
\Rightarrow gxH &= xH \\
\Rightarrow x^{-1}gxH &= H \quad \forall x \in G \\
\Rightarrow x^{-1}gx &\in H \quad \forall x \in G \\
\Rightarrow g &\in xHx^{-1} \quad \forall x \in G \\
\Rightarrow g &\in \bigcap_{x \in G} xHx^{-1}
\end{aligned}$$

Thus, $\ker \psi = \bigcap_{x \in G} xHx^{-1} \subseteq H$. Furthermore, $N = \ker \psi \triangleleft G$. By the First Isomorphism Theorem, $G/\ker \psi \cong \psi(G) \leq S_n$. Hence, $(G : N) = |G/N| = |\psi(G)| \leq |S_n| = n! < \infty$.

- b) Suppose $H_1, H_2 \leq G$ such that $(G : H_1) = n$, $(G : H_2) = m$. Consider the cosets of $H_1 \cap H_2$. Any left coset of $H_1 \cap H_2$ has the form $g(H_1 \cap H_2) = gH_1 \cap gH_2$ for $g \in G$. Hence, every left coset of $H_1 \cap H_2$ is the intersection of a left coset of H_1 with a left coset of H_2 . From $(G : H_1) = n$, $(G : H_2) = m$, there are at most nm distinct intersections and thus at most nm left cosets of $H_1 \cap H_2$. Hence, $(G : H_1 \cap H_2) \leq nm < \infty$. □

Exercise 0.8. (Exercise 10) Let G be a group and let H be a subgroup of finite index. Prove that there is only a finite number of right cosets of H , and that the number of right cosets is equal to the number of left cosets.

Proof. By assumption, $(G : H) = n < \infty$. The group G acts on G/H in two ways:

$$\lambda : G \rightarrow \text{Perm}(G/H)$$

by $x \mapsto x(gH)$ for some $g \in G$ and

$$\rho : G \rightarrow \text{Perm}(G/H)$$

by $x \mapsto (gH)x$ for some $g \in G$. Consider $eH = H \in G/H$. The isotropy group of eH in G is

$$G_{eH} = \{g \in G : g(eH) = eH = H\} = H.$$

The orbit of eH under G is

$$G \cdot eH = \{g(eH) : g \in G\} = G/H.$$

By the orbit-stabilizer theorem and Lagrange's Theorem,

$$|G/H| = |G \cdot eH| = \underbrace{(G : G_{eH})}_{\text{under } \lambda} = (G : H) = n < \infty.$$

Now, under ρ , $G_{eH} = \{g \in G : (eH)g = H\}$. If $Hg = H$, then $g = eg \in Hg = H$, so $g \in H$. Conversely, if $g \in G$, then clearly $Hg \subseteq H$ and $g^{-1} \in H$ gives $H \subseteq Hg$, so $H = Hg$. Hence, $G_{eH} = H$. The orbit of eH under ρ in G is

$$G \cdot (eH) = \{(eH)g : g \in G\} = Hg = H/G \quad (\text{notation in Lang for right cosets})$$

Again by the orbit-stabilizer theorem and Lagrange's

$$|H/G| = |G \cdot eH| = \underbrace{(G : G_{eH})}_{\text{under } \rho} = (G : H) = n < \infty.$$

Therefore, $|H/G| = |G/H| = n < \infty$, i.e. there is a finite number of right cosets and the number of left cosets is the same as the number of right cosets. □

Exercise 0.9. (Exercise 11) Let G be a group and A a normal abelian subgroup. Show that G/A operates on A by conjugation, and in this manner get a homomorphism of G/A into $\text{Aut}(A)$.

Proof. Define a map $\varphi : G/A \rightarrow \text{Aut}(A)$ by $\varphi(gA) = gAg^{-1}$, $g \in G$. (Well-defined) Suppose that $gA = hA$. Then $h = ga \in H$ for some $a \in A$. For any $x \in A$, we have $h x h^{-1} = (ga)x(ga)^{-1} = g(axa^{-1})g^{-1}$. Since A is abelian, $axa^{-1} = x$. Thus, $h x h^{-1} = g x g^{-1}$. Thus, the definition of φ does not depend on the choice of representative in G/A . ($\text{Im } \varphi \subseteq \text{Aut}(A)$) For each $g \in G$, the map $a \mapsto gag^{-1}$ is a homomorphism $A \rightarrow A$ since conjugation preserves products. It is bijective with inverse $a \mapsto g^{-1}ag$. Hence, $\varphi(gA) \in \text{Aut}(A)$. (Homomorphism) For $gA, hA \in G/A$ and $x \in A$,

$$\begin{aligned} (\varphi(gA) \circ \varphi(hA))(x) &= g(h x h^{-1})g^{-1} \\ &= (gh)x(gh)^{-1} = \varphi(ghA)(x). \end{aligned}$$

Therefore, φ is a group homomorphism. Thus, G/A acts on A by conjugation, and we obtain a homomorphism

$$\varphi : G/A \rightarrow \text{Aut}(A)$$

Its kernel is $\text{Ker } \varphi = C_G(A)/A$, where $C_G(A)$ is the centralizer of A in G . □

Semidirect Product

Exercise 0.10. (Exercise 12) Let G be a group and let H, N be subgroups with N normal. Let γ_x be conjugation by an element $x \in G$.

- Show that $x \mapsto \gamma_x$ induces a homomorphism $f : H \rightarrow \text{Aut}(N)$.
- If $H \cap N = \{e\}$, show that the map $H \times N \rightarrow HN$ given by $(x, y) \mapsto xy$ is a bijection, and that this map is an isomorphism if and only if f is trivial, i.e. $f(x) = \text{id}_N$ for all $x \in H$.
- We define G to be the **semidirect product** of H and N if $G = HN$ and $H \cap N = \{e\}$. Now, conversely, let N, H be groups, and let $\psi : H \rightarrow \text{Aut}(N)$ be a given homomorphism. Construct a semidirect product as follows. Let G be the set of pairs (x, h) with $x \in N$, $h \in H$. Define the composition law

$$(x_1, h_1)(x_2, h_2) = (x_1\psi(h_1)x_2, h_1h_2)$$

Show that this is a group law, and yields a semidirect product of N and H , identifying N with the set of elements $(x, 1)$ and H with the set of elements $(1, h)$.

Proof. a) First, let $e_H \in H$ be the identity element in H . Then, $f(e_H) = \gamma_{e_H}$. Let $n \in N$. Thus, $\gamma_{e_H}(n) = e_H n e_H^{-1} = n$ (since N normal). Thus, $f(e_H) = \gamma_{e_H} = \text{id}_N$. It is also well-defined since if $h_1 = h_2$, then $f(h_1) = \gamma_{h_1}(n) = h_1 n h_1^{-1}$ and $f(h_2) = \gamma_{h_2}(n) = h_2 n h_2^{-1}$. Clearly, we then have $f(h_1) = f(h_2)$. Let $x, y \in H$. Then, $f(xy) = \gamma_{xy}$. Take $n \in N$, $\gamma_{xy}(n) = x y n (xy)^{-1} = x y n y^{-1} x^{-1} = x(\gamma_y(n))x^{-1} = \gamma_x(\gamma_y(n)) = (\gamma_x \circ \gamma_y)(n) = f(x) \circ f(y)$ as desired.

- (Well-defined) For any $h \in H$, $n \in N$, $hn \in HN = \{xy : x \in H, y \in N\}$. (Injective) Suppose $(x_1, y_1), (x_2, y_2) \in H \times N$ and suppose $\varphi((x_1, y_1)) = \varphi((x_2, y_2)) \Rightarrow x_1 y_1 = x_2 y_2 \Rightarrow x_2^{-1} x_1 = y_2 y_1^{-1}$. Note $x_2^{-1} x_1 \in H$ and $y_2 y_1^{-1} \in N$. Thus, $x_2^{-1} x_1 \in H \cap N$ and $y_2 y_1^{-1} \in H \cap N$. But $H \cap N = \{e\}$ by assumption, so $x_2^{-1} x_1 = e \Rightarrow x_1 = x_2$ and $y_2 y_1^{-1} = e \Rightarrow y_1 = y_2$. Hence, $(x_1, y_1) = (x_2, y_2)$. (Surjective) Let $z \in HN$. Then, $z = hn$ for some $h \in H$, $n \in N$. Hence, $\varphi((h, n)) = hn = z$ with (h, n) clearly an element of $H \times N$. Thus, $\varphi : H \times N \rightarrow HN$ is a bijection.

Suppose $\varphi((x_1, y_1)(x_2, y_2)) = \varphi((x_1, y_1))\varphi((x_2, y_2))$ for $(x_i, y_i) \in H \times N$. This is if and only if

$$\begin{aligned}
x_1 x_2 y_1 y_2 &= x_1 y_1 x_2 y_2 \\
\iff x_2 y_1 &= y_1 x_2 \quad (\text{left/right cancellation}) \\
\iff y_1 &= x_2 y_1 x_2^{-1}
\end{aligned}$$

Since x_1, x_2, y_1, y_2 are arbitrary, this must hold for all $x_2 \in H$ and $y_1 \in N$. Hence,
 $\gamma_x|_N = \text{id}_N \Rightarrow f(x) = \text{id}_N$ for all $x \in H$.

- c) Let $\psi : H \rightarrow \text{Aut}(N)$ be given by $\psi(h) = \varphi_h$ with $\varphi_h(n) = n' \in N$. Thus, we define the composition of $(x_1, h_1), (x_2, h_2) \in N \times H$ by

$$\begin{aligned}
(x_1, h_1)(x_2, h_2) &= (x_1 \psi(h_1)(x_2), h_1 h_2) \\
&= (x_1 \varphi_{h_1}(x_2), h_1 h_2)
\end{aligned}$$

(Identity) Let $e = (e_N, e_H) \in G = N \times H$. Then,
 $(x_1, h_1)(e_N, e_H) = (x_1 \psi(h_1)(e_N), h_1 \cdot e_H) = (x_1 \varphi_{h_1}(e_N), h_1) = (x_1 \cdot e_N, h_1) = (x_1, h_1)$ (since $\varphi \in \text{Aut}(N)$ fixes the identity). Likewise, $(e_N, e_H)(x_1, h_1) = (x_1, h_1)$ and thus $e \in G$ which implies that $G \neq \emptyset$. (Closed) Observe that

$$\begin{aligned}
(x_1, h_1)(x_2, h_2) &= (x_1 \psi(h_1)(x_2), h_1 h_2) \\
&= (x_1 \varphi_{h_1}(x_2), h_1 h_2)
\end{aligned}$$

and since $\varphi_{h_1} \in \text{Aut}(N)$, $\varphi_{h_1}(x_2) \in N$. Hence, $x_1 \varphi_{h_1}(x_2) \in N$ and $h_1 h_2 \in H \Rightarrow (x_1 \varphi_{h_1}(x_2), h_1 h_2) \in G = N \times H$. (Associativity) Let $(x_1, h_1), (x_2, h_2), (x_3, h_3) \in G$. Then,

$$\begin{aligned}
(x_1, h_1) [(x_2, h_2)(x_3, h_3)] &= (x_1, h_1)(x_2 \psi(h_2)(x_3), h_2 h_3) \\
&= (x_1, h_1)(x_2 \varphi_{h_2}(x_3), h_2 h_3) = (x_1 \psi(h_1) [x_2 \varphi_{h_2}(x_3)], h_1 h_2 h_3) \\
&= (x_1 \varphi_{h_1}(x_2 \varphi_{h_2}(x_3)), h_1 h_2 h_3) \\
&= (x_1 \varphi_{h_1}(x_2) \varphi_{h_1}(\varphi_{h_2}(x_3)), h_1 h_2 h_3)
\end{aligned}$$

Now,

$$\begin{aligned}
[(x_1, h_1)(x_2, h_2)] (x_3, h_3) &= (x_1 \psi(h_1)(x_2), h_1 h_2)(x_3, h_3) \\
&= (x_1 \varphi_{h_1}(x_2), h_1 h_2)(x_3, h_3) \\
&= (x_1 \varphi_{h_1}(x_2) \psi(h_1 h_2)(x_3), h_1 h_2 h_3) \\
&= (x_1 \varphi_{h_1}(x_2) (\psi(h_1) \psi(h_2))(x_3), h_1 h_2 h_3) \\
&= (x_1 \varphi_{h_1}(x_2) \varphi_{h_1}(\varphi_{h_2}(x_3)), h_1 h_2 h_3)
\end{aligned}$$

Thus, $(x_1, h_1) [(x_2, h_2)(x_3, h_3)] = [(x_1, h_1)(x_2, h_2)] (x_3, h_3)$. (Inverses) Suppose that $(x_1, h_1)(x_2, h_2) = (e_N, e_H)$. Thus,

$$\begin{aligned}
(x_1, h_1)(x_2, h_2) &= (e_N, e_H) \\
\Rightarrow x_1 \psi(h_1)(x_2), h_1 h_2 &= (e_N, e_H) \\
\Rightarrow x_1 \varphi_{h_1}(x_2), h_1 h_2 &= (e_N, e_H) \\
\Rightarrow x_1 \varphi_{h_1}(x_2) &= e_N \quad \text{and} \quad h_1 h_2 = e_H \\
\Rightarrow \varphi_{h_1}(x_2) &= x_1^{-1} \quad \Rightarrow h_2 = h_1^{-1} \in H \\
\because \varphi_{h_1} &\in \text{Aut}(N), \varphi_{h_1^{-1}} \in \text{Aut}(N) \\
\Rightarrow \varphi_{h_1^{-1}}(\varphi_{h_1}(x_2)) &= \varphi_{h_1^{-1}}(x_1^{-1}) \Rightarrow x_2 = \varphi_{h_1^{-1}}(x_1^{-1}) \in N.
\end{aligned}$$

Therefore, $(x_1, h_1)^{-1} = (\varphi_{h_1^{-1}}(x_1^{-1}), h_1^{-1}) = (\psi(h_1^{-1})(x_1^{-1}), h_1^{-1}) \in G$. ($G = HN$) Let $(x, h) \in G = N \times H$. Then,

$$\begin{aligned}(x, h) &= (x\varphi_{e_H}(e_N), e_H \cdot h) \\ &= (x\psi(e_H)(e_N), e_H \cdot h) \\ &= \underbrace{(x, e_H)}_{\in N} \underbrace{(e_N, h)}_{\in H}\end{aligned}$$

Thus, $G = NH$. ($N \cap H = \{e\}$) Note that $(x, e_H) \in H \iff x \in e_N$ and $(e_N, h) \in N \iff h = e_H$. Thus, $N \cap H = \{(e_N, e_H)\} = \{e\}$. Therefore,

$$G = N \rtimes_{\psi} H.$$

□

Exercise 0.11. (Exercise 13)

- Let H, N be normal subgroups of a finite group G . Assume that the orders of H, N are relatively prime. Prove that $xy = yx$ for all $x \in H$ and $y \in N$, and that $H \times N \cong HN$.
- Let $H_1, \dots, H_r$ be normal subgroups of G such that the order of H_i is relatively prime to the order of H_j for $i \neq j$. Prove that

$$H_1 \times \dots \times H_r \cong H_1 \dots H_r.$$

Example. If the Sylow subgroups of a finite group are normal, then G is the direct product of its Sylow subgroups.

Proof. a) Assume $H, N \triangleleft G$ with $(|H|, |N|) = 1$. Consider the commutator,

$$[x, y] := xyx^{-1}y^{-1} \quad \forall x \in H, y \in N$$

Thus, $xyx^{-1}y^{-1} \in H$ since $yx^{-1}y^{-1} \in H$ by the fact that $H \triangleleft G$ and therefore $x(yx^{-1}y^{-1}) \in H$. Similarly $xyx^{-1}y^{-1} \in N$ since $xyx^{-1} \in N$ by the fact that $N \triangleleft G$. Thus, $[x, y] = xyx^{-1}y^{-1} \in H \cap N$. By Lagrange's Theorem,

$$|H \cap N| \mid |H| \quad \text{and} \quad |H \cap N| \mid |N|$$

and thus,

$$|H \cap N| \mid \underbrace{(|H|, |N|)}_{=1}$$

Hence, we must have $|H \cap N| = 1 \Rightarrow H \cap N = \{e\}$. Then since $[x, y] \in H \cap N$, it follows that $xyx^{-1}y^{-1} = e \Rightarrow xy = yx$. Finally, by *Exercise 12*, $H \times N \cong HN$.

- Assume $H_1, \dots, H_r \triangleleft G$ with $(|H_i|, |H_j|) = 1$ if $i \neq j$. From *part (a)*, for any H_i, H_j with $i \neq j$, $H_i \cap H_j = \{e\}$ and $xy = yx$ for all $x \in H_i, y \in H_j$. Hence, $H_i \times H_j \cong H_i H_j$. We proceed by induction. Suppose that $H_1 \times \dots \times H_{r-1} \cong H_1 \dots H_{r-1}$. Then, $(|H_r|, |H_1 \dots H_{r-1}|) = 1$ and $H_r \triangleleft G$. Thus, by the same argument as above, $H_1 \times \dots \times H_r \cong H_1 \dots H_r$.

□

Exercise 0.12. (Exercise 14)

Let G be a finite group and let N be a normal subgroup such that N and G/N have relatively prime orders.

- Let H be a subgroup of G having the same order as G/N . Prove that $G = HN$.

b) Let g be an automorphism of G . Prove that $g(N) = N$.

Proof. Let G be a finite group of order n and let $N \triangleleft G$ such that $(|N|, |G/N|) = 1$.

a) Assume that $H \leq G$ such that $|H| = |G/N| = (G : N)$. Then $(|H|, |N|) = 1$. We have

$$\begin{aligned} |HN| &= \frac{|H| \cdot |N|}{|H \cap N|} \\ &= \frac{|G/N| \cdot |N|}{|H \cap N|} \\ &= \frac{\frac{|G|}{|N|} \cdot |N|}{|H \cap N|} \\ &= \frac{|G|}{|H \cap N|} \end{aligned}$$

Then, if $|HN| = |G|$, we would require that $|H \cap N| = 1 \Rightarrow H \cap N = \{e\}$. Hence, since $HN \subseteq G$, we have that $G = HN$ if and only if $H \cap N = \{e\}$. Thus, we show the converse of this biconditional.

By Lagrange's Theorem, $|H \cap N| \mid |H|$ and $|H \cap N| \mid |N|$, and thus $|H \cap N| \mid \underbrace{(|H|, |N|)}_{=1}$. Hence,

$|H \cap N| = 1$ and therefore, $H \cap N = \{e\}$. It follows from the above biconditional that $G = HN$.

b) Let $g \in \text{Aut}(G)$. Then $g(N) \leq G$ with $|g(N)| = |N|$. By our assumption, $(|N|, |G/N|) = 1$, we have $(|g(N)|, |G/g(N)|) = 1$. Let $H \leq G$ such that $|H| = |G/N|$ as in *part (a)* which implies that $G = HN$ and $H \cap N = \{e\}$. Then $|g(H)| = |H| = |G/N|$ and applying *part (a)* to the normal subgroup $g(N)$ gives that $G = g(H) \cdot g(N)$ and $g(H) \cap g(N) = \{e\}$. Hence, since $N, g(N) \triangleleft G$ are the same order, admitting a complement of order $|G/N|$. The order of $Ng(N)$ is

$$|Ng(N)| = \frac{|N| \cdot |g(N)|}{|N \cap g(N)|} = \frac{|N|^2}{|N \cap g(N)|} \leq |G| = |N| \cdot |G/N|$$

Hence, we have that $|N \cap g(N)| \geq \frac{|N|}{|G/N|}$. Since $(|N|, |G/N|) = 1$, this implies that $|N \cap g(N)| = |N|$, so $N \subseteq g(N)$. By symmetry, using g^{-1} (since g is an automorphism), we have $g(N) \subseteq N$. Therefore, $g(N) = N$.

□

Some operations

Throughout, p is a prime number.

Exercise 0.13. (Exercise 20) Let P be a p -group. Let A be a normal subgroup of order p . Prove that A is contained in the center of P .

Proof. Assume $|A| = p$. Thus, $A \cong C_p$, the cyclic group of order p (since prime order implies cyclic). Thus, $\text{Aut}(A)$ is cyclic of order $p - 1$. Define

$$\varphi : P \rightarrow \text{Aut}(A)$$

by $g \mapsto c_g$, conjugation by g in A . Since $A \triangleleft P$, and by previous exercises, c_g is a bijective homomorphism of A into itself, hence $c_g \in \text{Aut}(A)$. Our map φ is a homomorphism since for $g, h \in P$ we have

$$\varphi(gh) = c_{gh} = c_g \circ c_h = \varphi(g)\varphi(h).$$

By definition, $\varphi(P) \leq \text{Aut}(A)$, thus $|\varphi(P)| \mid |\text{Aut}(A)| \Rightarrow |\varphi(P)| \mid p - 1$. But $\varphi(P)$ is also a homomorphic image of the p -group P , hence $|\varphi(P)| = p^k$ for some k . This is due to the First Isomorphism Theorem which states

$$P/\ker \varphi \cong \varphi(P).$$

Since $|P| = p^n$ for some n by assumption, and $\ker \varphi \leq P$, by Lagrange's Theorem, $|\ker \varphi| = p^k$ for $k \leq n$. Then, by counting sizes via the isomorphism,

$$|\varphi(P)| = |P/\ker \varphi| = \frac{|P|}{|\ker \varphi|} = \frac{p^n}{p^k} = p^{n-k} \quad (\text{a power of } p)$$

Whence, the only positive integer that is both a power of p and a divisor $p-1$ is 1. Thus, $|\varphi(P)| = p^0 = 1$. This means $\varphi(P)$ is trivial, i.e. $\varphi(g) = \text{id}_A$ for all $g \in P$. Then, $(\varphi(g))(a) = c_g(a) = gag^{-1} = a$ (since $\varphi(g)(a) = a$ by $\varphi(g) = \text{id}_A$). Hence, $ga = ag$. Therefore every element of A commutes with every element of P . Therefore,

$$A \subseteq Z(P).$$

□

Exercise 0.14. (Exercise 21) Let G be a finite group and H a subgroup. Let P_H be a p -Sylow subgroup of H . Prove that there exists a p -Sylow subgroup P of G such that $P_H = P \cap H$.

Proof. Let G be a finite group, $H \leq G$, and let P_H be a p -Sylow subgroup of H . By Theorem 6.4, there exists a p -Sylow subgroup $P \leq G$ such that every p -subgroup of G is contained in P . Hence, since $H \leq G$, P_H is a p -subgroup of G (p -Sylow subgroup of H , by assumption) and thus

$$P_H \subseteq P$$

Furthermore, $P_H \subseteq H \Rightarrow P_H \subseteq P \cap H$. Observe that $P \cap H \leq H$ and thus since P is a p -subgroup (also Sylow), $P \cap H$ is a p -subgroup of H and therefore $P \cap H \subseteq P_H$. Hence,

$$P \cap H = P_H.$$

□

Exercise 0.15. (Exercise 22)

Let H be a normal subgroup of a finite group G and assume that $\#(H) = p$. Prove that H is contained in every p -Sylow subgroup of G .

Proof. Assume that $H \triangleleft G$ and suppose $|H| = p$. This follows nicely from all parts of Theorem 6.4. Thus, H is a p -subgroup and for all $x \in G$,

$$xHx^{-1} = H \quad (\text{normality})$$

Then by definition, $N(H) = G$. By Theorem 6.4(i), since H is a p -subgroup of G , H is contained in some p -Sylow subgroup, say K . Then, by Theorem 6.4(ii), all p -Sylow subgroups are conjugate and thus $K' = gKg^{-1}$ is a p -Sylow subgroup for some $g \in G$. Conjugating the inclusion $H \subseteq K$, we get

$$H = xHx^{-1} \subseteq xKx^{-1} = K'$$

Hence, $H \subseteq K'$ with K' a different, unless K normal, p -Sylow subgroup. Thus, we can continue this process until we get all $pk+1$ p -Sylow subgroups ($k \in \mathbb{Z}$, by Theorem 6.4(iii) which states the number of p -Sylow subgroups of G is $\equiv 1 \pmod{p}$). Thus, it follows that H is contained in every p -Sylow subgroup.

□

Exercise 0.16. (Exercise 23) Let P, P' be p -Sylow subgroups of a finite group G .

- a) If $P' \subseteq N(P)$ (normalizer of P), then $P' = P$.
- b) If $N(P') = N(P)$, then $P' = P$.
- c) We have $N(N(P)) = N(P)$.

Proof. a) Suppose $P' \subseteq N(P)$. Then for all $x \in P'$, we have

$$xPx^{-1} = P$$

and thus $P \triangleleft P'$. Since P, P' are p -groups and P is a normal subgroup of P' , the quotient P'/P is also a p -group. By definition, $|P| = |P'| = p^n$ with p^n the highest power of p dividing $|G|$. Hence, $|P'/P| = \frac{|P'|}{|P|} = 1$. Thus, $P = P'$.

b) We know $P' \subseteq N(P')$ and $P \subseteq N(P)$. By assumption $N(P') = N(P)$, so

$$P' \subseteq N(P') = N(P)$$

Thus, by part (a), $P' = P$.

c) $(N(P) \subseteq N(N(P)))$ This inclusion follows trivially from the fact that for any subgroup $H \leq G$, $H \subseteq N(H)$, i.e. every subgroup is a subset of its normalizer. Thus, if you let $K = N(H)$, then we have $K \subseteq N(K)$.

$(N(N(P)) \subseteq N(P))$ Let $x \in N(N(P))$. Then

$$xN(P)x^{-1} = N(P)$$

and in particular,

$$xPx^{-1} \subseteq N(P)$$

Thus, xPx^{-1} is a p -Sylow subgroup of $N(P)$. Since $P \subseteq N(P)$, P is also a p -Sylow subgroup of $N(P)$. By Theorem 6.4, there exists $y \in N(P)$ such that

$$y(xPx^{-1})y^{-1} = P$$

Then $yx \in N(P)$ and since $N(P)$ is a subgroup, we have that $x \in N(P)$ as was to be shown. Therefore, $N(N(P)) = N(P)$. □

Explicit determination of groups

Exercise 0.17. (Exercise 24)

Let p be a prime number. Show that a group of order p^2 is abelian, and that there are only two such groups up to isomorphism.

Proof. Suppose $|G| = p^2$ for a prime p . Thus, G is a p -group and hence $Z(G)$ is not trivial. Recall that the class equation is defined as

$$|G| = |Z(G)| + \sum_{x \in C} (G : G_x)$$

where C is a set of representatives for the distinct conjugacy classes and G_x is the isotropy group of x in G defined as $G_x = \{x \in G : xy = y\}$ with $y \in G$ (here G is acting on itself via conjugation). Each term in $\sum_{x \in C} (G : G_x) > 1$ and divides $|G| = p^2$. Hence, $\sum_{x \in C} (G : G_x) \equiv 0 \pmod{p}$. Thus,

$$|G| = |Z(G)| + kp \quad (k \in \mathbb{Z}^+) \quad \left[kp = \sum_{x \in C} (G : G_x) \right]. \text{ Therefore, } |Z(G)| = |G| - kp \Rightarrow |Z(G)| \equiv 0 \pmod{p}.$$

Thus, $p \mid |Z(G)| \Rightarrow |Z(G)| = p$ or p^2 .

But by **Exercise 1**, Cases 1 and 2 cover this, thus, it follows that G is abelian.

If G has an element g of order p^2 , then the order of $\langle g \rangle$ is p^2 and thus $G = \langle g \rangle$. Therefore, in this case, $G \cong \mathbb{Z}/p^2\mathbb{Z}$.

If no element (not including the identity) of G has order p^2 , then it must have order p . Choose $a \in G$ such that $|\langle a \rangle| = p$ and thus $(G : \langle a \rangle) = p$. Choose $b \notin \langle a \rangle$. Hence, we also have $|b| = p$. Hence, we have $\langle a \rangle \cap \langle b \rangle = \{e\}$ with the product $\langle a \rangle \times \langle b \rangle$ having order p^2 . Therefore, $G = \langle a \rangle \times \langle b \rangle \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

Then these two possibilities of G can not be isomorphic since one is cyclic and one is not, and cyclicity is invariant under isomorphisms. □

Exercise 0.18. (Exercise 25)

Let G be a group of order p^3 , where p is prime, and G is not abelian. Let Z be its center. Let C be a cyclic group of order p .

- a) Show that $Z \cong C$ and $G/Z \cong C \times C$.
- b) Every subgroup H of order p^2 contains Z and is normal.
- c) Suppose $x^p = 1$ for all $x \in G$. Show that G contains a normal subgroup $H \cong C \times C$.

Proof. a) Assume that G is not an abelian group with order p^3 . Since $|G| = p^3$, G is a p -group. By Theorem 6.5, since $p^3 > 1$, G has a nontrivial center. By the assumption that G is not abelian, $G \neq Z$. Thus, $|Z| = p$ or p^2 . If $|Z| = p^2$, then $|G/Z| = p$ and thus G/Z is cyclic. By **Exercise 1**, it follows that G is abelian, a contradiction! Therefore $|Z| = p$ and thus Z is cyclic of order p , i.e. $Z \cong C$. Then $|G/Z| = p^2$. By **Exercise 24**, there are two abelian groups, up to isomorphism, of order p^2 , a cyclic group and a non-cyclic group. If G/Z is cyclic, then by **Exercise 1**, G is abelian, a contradiction! Hence, G/Z is not cyclic which implies that $G/Z \cong C \times C$.

- b) Let $H \leq G$ such that $|H| = p^2$. Thus, by **Exercise 24**, H is abelian. It follows that $H \cong C_{p^2} \cong \mathbb{Z}/p^2\mathbb{Z}$ or $H \cong C \times C$. From part (a), $|Z| = p$. Hence, $|G/Z| = p^2$. In part (a), we showed that

$$G/Z \cong C \times C.$$

Consider

$$\pi: G \rightarrow G/Z$$

given by $g \mapsto gZ$. Thus

$$\ker \pi = \{x \in G: xZ = Z\} = G \cap Z = Z$$

Let $\pi|_H$ be the restriction to H . Then

$$\ker \pi|_H = H \cap Z \subseteq Z.$$

Since $\ker \pi|_H \triangleleft H$, Lagrange's and the containment, $|H \cap Z| \mid |Z| \Rightarrow |H \cap Z| = 1$ or p . By the First Isomorphism Theorem,

$$\begin{aligned} H / \ker \pi|_H &\cong \pi(H) \\ \Rightarrow H / H \cap Z &\cong \pi(H) \end{aligned}$$

Therefore,

$$|H / H \cap Z| = |\pi(H)| \Rightarrow \frac{|H|}{|H \cap Z|} = |\pi(H)|.$$

If $|H \cap Z| = 1$, then $|\pi(H)| = |H| = p^2 = |G/Z|$. This would mean that H maps surjectively onto G/Z but $H \leq G$. Thus, we must have $|H \cap Z| = p$. Hence since $Z \cong C$, we have $H \cap Z = Z$ and therefore $Z \subseteq H$. Then $|\pi(H)| = p$. Thus, $\pi(H) \leq G/Z \cong C \times C$. Since G/Z is abelian, $\pi(H) \triangleleft G/Z$. The pre-image of a normal subgroup is normal, hence $\pi^{-1}(\pi(H)) \triangleleft G$. The pre-image is defined as

$$\begin{aligned}\pi^{-1}(\pi(G)) &= \{x \in G : xZ \in \pi(H)\} \\ &= \{x \in G : xZ = yZ \text{ for some } h \in H\}\end{aligned}$$

Let $g \in \pi^{-1}(\pi(H))$. Thus, $\pi(g) \in \pi(H)$ which implies that $\pi(g) = \pi(h)$ for some $h \in H$. Then $gZ = hZ$. Hence, there exists some $x \in Z$ such that $g = hx$. Since $Z \subseteq H$, we have $g = hx \in H$. Hence, $\pi^{-1}(\pi(H)) \subseteq H$. Now let $h \in H$. Then $\pi(h) = hZ \in \pi(H)$. Thus, $H \subseteq \pi^{-1}(\pi(H))$. Therefore, $\pi^{-1}(\pi(H)) = H$. Then from

$$\pi^{-1}(\pi(H)) \triangleleft G,$$

we have $H \triangleleft G$. Therefore, $Z \subseteq H$ and H is normal in G .

c) Since $G/Z \cong C \times C$ from part (a), G/Z contains a cyclic subgroup of order p , say

$$\langle aZ \rangle \leq G/Z.$$

Let

$$\begin{aligned}H &= \pi^{-1}(\langle aZ \rangle) = \{x \in G : \pi(x) \in \langle aZ \rangle\} \\ &= \{x \in G : xZ = a^k Z, k \in \mathbb{Z}\} \\ &= \{x \in G : x = a^k z, k \in \mathbb{Z}, z \in Z\}.\end{aligned}$$

Thus, $|H| = |\langle aZ \rangle| \cdot |Z| = p^2$. By part (b), $H \triangleleft G$ and $Z \subseteq H$. By assumption, $x^p = 1$ for all $x \in G$, so all elements of H have orders dividing p . Therefore, since $|H| = p^2$ and no element of H has order p^2 ($p^2 \nmid p$), H cannot be cyclic. Thus, by **Exercise 24**, $H \cong C \times C$. □

Exercise 0.19. (Exercise 26)

- a) Let G be a group of order pq , where p, q are primes and $p < q$. Assume that $q \not\equiv 1 \pmod{p}$. Prove that G is cyclic.
- b) Show that every group of order 15 is cyclic.

Proof. a) Let $|G| = pq$ with $p < q$ distinct primes and assume that $q \not\equiv 1 \pmod{p}$. Thus, there exists p, q -Sylow subgroups H_p, H_q respectively with $|H_p| = p$ and $|H_q| = q$. By Lemma 6.7, since $(G : H_q) = p < q$ is the smallest prime dividing the order of G , we have $H_q \triangleleft G$. Furthermore, since H_p, H_q are both prime order, they are cyclic (Proposition 4.1) and thus there exists $x \in H_p$ and $y \in H_q$ such that $H_p = \langle x \rangle$ and $H_q = \langle y \rangle$. Define

$$\varphi : H_p \rightarrow \text{Aut}(H_q)$$

by $x \mapsto \varphi_x$ with $\varphi_x : H_q \rightarrow H_q$ being conjugation by x , an automorphism. Since H_q is cyclic, any automorphism must send a generator to another generator, i.e. $y \mapsto y^k$. Thus, $k \in (\mathbb{Z}/q\mathbb{Z})^\times$. Then

$$\text{Aut}(H_q) \cong (\mathbb{Z}/q\mathbb{Z})^\times$$

which implies that $|\text{Aut}(H_q)| = q - 1$. Then

$$\begin{aligned}\ker \varphi &= \{x \in H_p : \varphi_x(y) = y, \forall y \in H_q\} \\ &= \{x \in H_p : xy = yx, \forall y \in H_q\} \triangleleft H_p\end{aligned}$$

By Lagrange's Theorem, $|\ker \varphi| \mid |H_p|$. Hence $|\ker \varphi| = 1$ or p . If $|\ker \varphi| = 1$, by the First Isomorphism Theorem, we have

$$H_p / \ker \varphi \cong \varphi(H_p) \leq \text{Aut}(H_q)$$

and thus $|H_p / \ker \varphi| = |H_p| = p = |\varphi(H_p)|$. Then, by Lagrange's Theorem, $|\varphi(H_p)| \mid |\text{Aut}(H_q)|$, but by assumption, $q \not\equiv 1 \pmod p$. Hence we can not have $|\ker \varphi| = 1$. Thus, $|\ker \varphi| = p$ and therefore by the First Isomorphism Theorem, $|H_p / \ker \varphi| = 1 = |\varphi(H_p)|$. Thus, $\varphi(H_p) = \{\text{id}\}$. Since φ is trivial, every element of H_p commutes with every element of H_q . Thus, $H_p H_q$ is abelian. Let $z \in H_p \cap H_q$. Then $z \in H_p$ and $z \in H_q$. Hence, $z = x^k$ and $z = y^l$ for some $k \in (\mathbb{Z}/p\mathbb{Z})^\times$ and $l \in (\mathbb{Z}/q\mathbb{Z})^\times$. Then

$$\begin{aligned} x^k &= y^l \\ (x^k)^p &= (y^l)^p \\ e &= y^{lp} \end{aligned}$$

Thus, $q \mid lp$. By Euclid's Lemma, since q is a prime and $q \nmid p$, we have $q \mid l$ which implies that $l \equiv 0 \pmod q$ and $y^l = e$. Then $x^k = y^l = e \Rightarrow k \equiv 0 \pmod p$ and therefore $z = e$. Hence $H_p \cap H_q = \{e\}$. Thus,

$$|H_p H_q| = \frac{|H_p| \cdot |H_q|}{|H_p \cap H_q|} = \frac{pq}{1} = pq = |G|$$

Then $G = H_p H_q$. Since H_p, H_q are cyclic with $(p, q) = 1$, H_p, H_q abelian, and $H_p \cap H_q = \{e\}$, we have $H_p H_q = \langle xy \rangle$ with $|xy| = \text{lcm}(|x|, |y|) = \text{lcm}(p, q) = pq = |G|$. Therefore, $\langle xy \rangle$ has order pq and thus $G = H_p H_q = \langle xy \rangle$ is cyclic.

b) Let $|G| = 15 = 3 \cdot 5$. Since $5 \not\equiv 1 \pmod 3$, by part (a), G is cyclic. □

Exercise 0.20. (Exercise 28) Let p, q be distinct primes. Prove that a group of order $p^2 q$ is solvable, and that one of its Sylow subgroups is normal.

Proof. Let n_p and n_q denote the number of p -Sylow and q -Sylow subgroups G , respectively. By Sylow's Theorems,

$$n_p \mid q, \quad n_p \equiv 1 \pmod p, \quad n_q \mid p^2, \quad n_q \equiv 1 \pmod q$$

Hence,

$$n_p \in \{1, q\}, \quad n_q \in \{1, p, p^2\}$$

Case 1: ($p < q$) If $n_q = p$, then $q \mid p - 1$ which is impossible since $q > p - 1$. If $n_q = p^2$, then $q \mid (p - 1)(p + 1)$ which is impossible since $q > p + 1$ (as well). Thus, $n_q = 1$, so the q -Sylow subgroup $Q \triangleleft G$ is normal. Case 2: ($p > q$) If $n_p = q$, then $p \mid q - 1$ which is impossible since $p > q - 1$. Thus, $n_p = 1$, so the p -Sylow subgroups $P \triangleleft G$ is normal. In both cases, G has a normal Sylow subgroup N (either $|N| = q$ or $|N| = p^2$). Now consider the quotient G/N .

- $|N|$ is a prime power and therefore is solvable.
- $|G/N|$ is either p^2 or pq and in either case, both are known to be solvable.

Since both N and G/N are solvable, it follows that G is solvable. [The fact that $n_p, n_q = 1 \Rightarrow P, Q$ are normal is since all Sylow groups are conjugate and if there is only one Sylow subgroup, then it is fixed by conjugation by G , hence normal.] □

Exercise 0.21. (Exercise 29)

Let p, q be odd primes. Prove that a group of order $2pq$ is solvable.

Proof. Let G be a group of order $2pq$ and without loss of generality suppose that $q < p$. Thus, we have $2 < q < p$. We need a lemma.

Lemma 0.1. If p, q, r are any three prime numbers, then a group of order pqr is not simple.

Proof. Recall that a group is said to be *simple* if the group has no nontrivial normal subgroups (i.e. the trivial subgroup and the group itself are the only normal subgroups).

Case 1: $p = q = r$: If $p = q = r$, then $|G| = p^3$. By **Exercise 25**, we see at once that G contains a nontrivial normal subgroup and thus is not simple.

Case 2: $p = q \neq r$: If $p = q \neq r$, then $|G| = p^2r$. Let n_p denote the number of p -Sylow subgroups of G and n_r denote the number of r -Sylow subgroups of G . Thus, by Theorem 6.4,

$$n_p \mid r, n_p \equiv 1 \pmod{p} \quad \text{and} \quad n_r \mid p^2, n_r \equiv 1 \pmod{r}$$

Thus, $n_p \in \{1, r\}$. If $n_p = 1$, let P denote the unique p -Sylow subgroup of G . Observe that since G is not a p -group, $P \neq G$. Then since p -Sylow subgroups are conjugate and $n_p = 1$, it follows that for any $g \in G$, $gPg^{-1} = P$. Hence, $P \triangleleft G$. Therefore G is not simple. Hence, assume that $n_p = r$. Thus, $r \equiv 1 \pmod{p}$, so $r > p$. Since $n_r \mid p^2$ and $n_r \equiv 1 \pmod{r}$, $n_r \in \{1, p, p^2\}$. If $n_r = 1$, then by the same logic as in the case for $n_p = 1$, G has a normal r -Sylow subgroup and therefore G is not simple. Therefore, suppose that $n_r \neq 1$. If $n_r = p$, then $p \equiv 1 \pmod{r}$ which implies that $p > r$, a contradiction! Thus, we must have $n_r = p^2$. But each pair of distinct r -Sylow subgroups intersect trivially, so the total number of non-identity elements contained in all r -Sylow subgroups is

$$p^2(r - 1).$$

Including the identity, this accounts for $p^2(r - 1) + 1$ elements of G . Since $|G| = p^2r$, the number of elements not contained in any r -Sylow subgroup is

$$p^2r - (p^2(r - 1) + 1) = p^2 - 1.$$

Each p -Sylow subgroup contains $p^2 - 1$ non-identity elements, and distinct p -Sylow subgroups intersect trivially. Hence there is room for at most one p -Sylow subgroup, contradicting the assumption that $n_p = r > 1$.

Case 3: p, q, r distinct: Assume that p, q, r are distinct primes and without loss of generality assume that $r < q < p$. As previously stated, if any of n_p, n_q, n_r are equal to 1, then there is a normal p, q , or r -Sylow subgroup and we are done. Thus suppose this is not the case, i.e. $n_p, n_q, n_r \neq 1$. Since p is the largest prime, n_p cannot be equal to q or r . Hence $n_p = qr$. Furthermore, since $q > r$, $n_q \neq r$ and hence $n_q \geq p$. And finally, $n_r \geq q$. Since distinct p -Sylow subgroups intersect trivially, the number of non-identity elements contained in all p -Sylow subgroups is

$$n_p(p - 1) = qr(p - 1).$$

Including the identity, this accounts for $qr(p - 1) + 1$ elements of G . Since $|G| = pqr$, the number of elements not contained in any p -Sylow subgroup is

$$pqr - (qr(p - 1) + 1) = qr - 1.$$

Each q -Sylow subgroup contains $q - 1$ non-identity elements, and distinct q -Sylow subgroups intersect trivially. Hence the total number of non-identity elements contained in all q -Sylow subgroups is at least

$$n_q(q - 1) \geq p(q - 1).$$

But since $p > r$, we have

$$p(q - 1) > r(q - 1) = qr - r < qr - 1,$$

which is impossible since there are only $qr - 1$ elements available outside the p -Sylow subgroups. This contradiction completes the proof. □

Returning to the proof of the exercise, by the Lemma we just proved, G is not simple. Thus, there exists a normal subgroup $N \triangleleft G$ with $N \neq \{e\}$ and $N \neq G$. Hence N either has prime order or has order of a product of two distinct primes. Similarly, the quotient group G/N has the same possible orders depending on the order of N . From the proof of Exercise 28, it follows that N and G/N are solvable, which implies that G is solvable. □

Exercise 0.22. (Exercise 30)

- a) Prove that one of the Sylow subgroups of a group of order 40 is normal.
- b) Prove that one of the Sylow subgroups of a group of order 12 is normal.

Proof. a) Let G be a group of order $40 = 2^3 \cdot 5$. Let $P = \text{Syl}_2(G)$, $Q = \text{Syl}_5(G) \leq G$. Denote by n_2 and n_5 the number of 2-, 5-Sylow subgroups of G . By Theorem 6.4,

$$n_2 \equiv 1 \pmod{2}, n_2 \mid 5 \quad \text{and} \quad n_5 \equiv 1 \pmod{5}, n_5 \mid 8.$$

Hence, $n_2 \in \{1 + 2k : k \in \mathbb{Z}_{\geq 0}\} \cap \{x \in \mathbb{Z}^+ : x \mid 5\} = \{1, 5\}$. From this alone, we cannot determine if G contains a normal 2-Sylow subgroup or not. Thus, we see if G contains a normal 5-Sylow subgroup. We have $n_5 \in \{1 + 5k : k \in \mathbb{Z}_{\geq 0}\} \cap \{x \in \mathbb{Z}^+ : x \mid 8\} = \{1\}$. Hence, $n_5 = 1$. Again, by Theorem 6.4, since all 5-Sylow subgroups are conjugate and there is only one 5-Sylow subgroup Q , we deduce that

$$gQg^{-1} = Q \quad \forall g \in G.$$

Hence, $Q \triangleleft G$.

- b) Let G be a group of order $12 = 2^2 \cdot 3$. Let $P = \text{Syl}_2(G)$, $Q = \text{Syl}_3(G) \leq G$. Then, by the using the same notation and logic of part (a),

$$n_2 \equiv 1 \pmod{2}, n_2 \mid 3 \quad \text{and} \quad n_3 \equiv 1 \pmod{3}, n_3 \mid 4.$$

Then $n_2 \in \{1 + 2k : k \in \mathbb{Z}_{\geq 0}\} \cap \{x \in \mathbb{Z}^+ : x \mid 3\} = \{1, 3\}$ and $n_3 \in \{1 + 3k : k \in \mathbb{Z}_{\geq 0}\} \cap \{x \in \mathbb{Z}^+ : x \mid 4\} = \{1, 4\}$. If $n_3 = 1$, then we are done as we would deduce that $Q \triangleleft G$. Thus, assume that $n_3 = 4$. Hence, there are 4 3-Sylow subgroups. These intersect trivially, so in total they contribute $4 \cdot (3 - 1) = 8$ non-identity elements to G . Observe that it is impossible to also have that $n_2 = 3$. If this was the case, then the 3 2-Sylow subgroups would contribute $3 \cdot (2 - 1) = 3$ non-identity elements to G which contradicts $|G| = 12 \neq 8 + 3 + 1 = 12$. Therefore, we must have that $n_2 = 1$ and since all 2-Sylow subgroups are conjugate, and P is the only 2-Sylow subgroup, we have

$$gPg^{-1} = P \quad \forall g \in G.$$

Hence, $P \triangleleft G$. □

Exercise 0.23. (Exercise 32) Let S_n be the permutation group on n elements. Determine the p -Sylow subgroups of S_3 , S_4 , S_5 for $p = 2$ and $p = 3$.

Proof. We utilize Thm 6.2 throughout the following derivations to justify existence. We will find the Sylow subgroups up to isomorphism. (S_3 ; order $3! = 6 = 2 \cdot 3$) Let $\mathbf{p} = \mathbf{2}$. The highest power of 2 dividing 6 is 2. Hence, let $P_2 \leq S_3$ be a 2-Sylow subgroups and hence $|P_2| = 2$. Thus, $P_2 \cong C_2$ (P_2 is the subgroup generated by any of the three transpositions of S_3). Let $\mathbf{p} = \mathbf{3}$. The highest power of 3 dividing 6 is 3. Hence, let $P_3 \leq S_3$ be a 3-Sylow subgroup, $|P_3| = 3$. Observe $(S_3 : P_3) = 2$, thus $P_3 \triangleleft S_3$ and the only nontrivial normal subgroup of S_3 is A_3 . Hence, $P_3 = A_3 = \langle (123) \rangle \cong C_3$. (S_4 ; order $4! = 24 = 2^3 \cdot 3$) Let $\mathbf{p} = \mathbf{2}$. A 2-Sylow subgroup P_2 has order $2^3 = 8$. Note that all elements of a 2-Sylow subgroup have an order of a power of 2. For example:

- Transpositions have order 2.

- The product of disjoint transpositions have order 2.
- 4-cycles have order 4.

Thus, there are no 3-cycles. We will construct a 2-Sylow subgroup. Start with disjoint the disjoint transpositions $(12), (34)$. These cycles generate the Klein 4-group,

$$V_4 = \{(1), (12), (34), (12)(34)\}$$

Thus far, we have order 4. Now consider the permutation $(13)(24)$. Conjugation by this element yields,

$$(13)(24)(12)(13)(24) = (43) = (34)$$

Hence, $(13)(24)$ swaps the two generators of V_4 . Now the subgroup

$$P_2 = \langle (12), (34), (13)(24) \rangle$$

contains V_4 as a normal subgroup and an element acting by symmetry. Furthermore, $|P_2| = 8$. A group of order 8 containing a normal Klein 4-subgroup, plus an element that swaps its generators, is exactly the Dihedral group, D_4 . Hence,

$$P_2 \cong D_4$$

Thus, all 2-Sylow subgroups of S_4 are conjugate and isomorphic to D_4 . Let $\mathbf{p} = \mathbf{3}$. A 3-Sylow subgroup of S_4 has order 3 and thus is generated by any 3-cycle. Thus, $P_3 \cong C_3$. S_5 ; order $5! = 120 = 2^3 \cdot 3 \cdot 5$
Since the 2-,3-Sylow subgroups of S_5 will have the same order as the 2-,3-Sylow subgroups of S_4 , we have $P_2 \cong D_4$ and $P_3 \cong C_3$. □

Exercise 0.24. (Exercise 35)

Show that there are exactly two non-isomorphic non-abelian groups of order 8. (One of them is given by generators σ, τ with the relations

$$\sigma^4 = 1, \quad \tau^2 = 1, \quad \tau\sigma\tau = \sigma^3.$$

The other is the quaternion group.)

Proof. Assume that G is a non-abelian group of order 8. Then there are no elements in G of order 8 since that would imply that G is cyclic and therefore abelian. Hence every non-identity element of G has order 2 or 4. If every element has order 2, then if $x, y \in G$, we have

$$\begin{aligned} \underbrace{(xy)}_{\in G} &= e \\ xyxy &= e \\ xy &= y^{-1}x^{-1} \\ xy &= yx \quad \because y^{-1} = y, x^{-1} = x \end{aligned}$$

But then G is abelian. Thus there exists at least one $g \in G$ of order 4. Hence, $(G : \langle g \rangle) = 2$ which implies that $\langle g \rangle \triangleleft G$. Then

$$hgh^{-1} \in \langle g \rangle$$

for all $h \in G$. Then since hgh^{-1} has order 4, we have $hgh^{-1} = g$ or g^{-1} . In the first case, $hg = gh$ which implies that G is abelian. Thus $hgh^{-1} = g^{-1}$. The quotient group $G/\langle g \rangle$ has order 2 and thus for every $h \in G/\langle g \rangle$, $h^2 = e \in G/\langle g \rangle \Rightarrow h^2 \in \langle g \rangle$. By construction of G , $|h| = 2$ or 4 , so $|h^2| = 1$ or 2 . Therefore,

$$h^2 \in \{e, g^2\}.$$

If $h^2 = e$, then $G = \langle g, h \rangle$ with $g^4 = e$, and $hgh^{-1} = g^{-1} = g^3$ which is precisely the definition of D_4 . For the other case, $h^2 = g^2$, $g^4 = e$, and $hgh^{-1} = g^{-1}$ and this is precisely the quaternion group. □

Exercise 0.25. (Exercise 36)

Let $\sigma = [123 \cdots n]$ in S_n . Show that the conjugacy class of σ has $(n-1)!$ -elements. Show that the centralizer of σ is the cyclic group generated by σ .

Proof. Recall the class equation is given by

$$(G:1) = |G| = \sum_{x \in C} (G:G_x) = \sum_{x \in C} |\bar{x}|$$

with C a set of distinct representatives for each conjugacy class, G_x the isotropy group of x in G , and $\bar{x}$ denoting the conjugacy class of x . Hence, for this exercise, we consider $G = S_n$ acting on itself via conjugation and therefore $G_x = Z(x)$, the centralizer of x . Then we have

$$|S_n| = \sum_{\tau \in C} (S_n:Z(\tau)) = \sum_{\tau \in C} |\bar{\tau}|.$$

Therefore, since the sums are indexed by the same set C and are equal, we have

$$|\bar{\sigma}| = (S_n:Z(\sigma)) = \frac{|S_n|}{|Z(\sigma)|}$$

with $\sigma = (123 \cdots n) \in S_n$ (we use parentheses to denote a permutation instead of brackets, as it is given in the exercise statement). It should be noted that, by definition, $|\sigma| = n$. Thus, if we can show that $Z(\sigma) = \langle \sigma \rangle$, our results will follow.

Let $\tau \in Z(\sigma)$. Thus $\tau\sigma = \sigma\tau$. Evaluating at 1,

$$(\tau\sigma)(1) = \tau(\sigma(1)) = \tau(2) = \sigma(\tau(1)).$$

Then, if we know what $\tau(1)$ is equal to, we can then compute $\tau(k)$ for any $k \in \{1, \dots, n\}$. Suppose $\tau(1) = j$ for some $j \in \{1, \dots, n\}$. Then,

$$\begin{aligned} \tau(2) &= \sigma(\tau(1)) = \sigma(j) \\ \tau(3) &= \sigma(\tau(2)) = \sigma^2(j) \\ &\dots \\ \tau(k) &= \sigma^{k-1}(j), \quad 1 \leq k \leq n \end{aligned}$$

Furthermore, $\sigma^l(j) = j + l \pmod n$. Hence,

$$\begin{aligned} \tau(1) &= j \in \{1, \dots, n\} \\ \tau(2) &= \sigma(j) = j + 1 \pmod n \\ &\dots \\ \tau(n) &= \sigma^{n-1}(j) = j + (n-1) \pmod n \end{aligned}$$

For τ to be a permutation, these must be distinct. Thus, there are exactly n permutations that commute with σ . Then,

$$\begin{aligned} \tau(1) &= \sigma^0 = \text{id}. \\ \tau(2) &= \sigma^1 \\ &\vdots \\ \tau(n) &= \sigma^{n-1} \end{aligned}$$

are distinct. Hence

$$\begin{aligned} Z(\sigma) &= \{\tau \in S_n: \tau\sigma = \sigma\tau\} \\ &= \{\sigma^k: 0 \leq k < n\} \\ &= \langle \sigma \rangle. \end{aligned}$$

Therefore $Z(\sigma)$ is cyclic generated by σ and $|Z(\sigma)| = n = |\sigma|$. Then by the equality

$$|\bar{\sigma}| = \frac{|S_n|}{|Z(\sigma)|},$$

it follows that

$$|\bar{\sigma}| = \frac{n!}{n} = (n-1)!$$

Therefore, there are $(n-1)$ -elements in the conjugacy class of σ . □

Exercise 0.26. (Exercise 39)

Show that the action of the alternating group A_n on $\{1, \dots, n\}$ is $(n-2)$ -transitive.

Proof. Let $J_n = \{1, \dots, n\}$. First, note that A_n acts transitively (1-transitive) on J_n since $A_n \cdot 1 = \{\gamma \cdot 1 : \gamma \in A_n\} = J_n$, i.e. only one orbit. Let $X_{n-2} = \{x_1, \dots, x_{n-2}\}$, $Y_{n-2} = \{y_1, \dots, y_{n-2}\}$ and set $A = J_n \setminus X_{n-2} = \{x_{n-1}, x_n\}$ and $B = J_n \setminus Y_{n-2} = \{y_{n-1}, y_n\}$. Since S_n is n -transitive, and thus $(n-2)$ -transitive, there exists some $\gamma \in S_n$ such that $\gamma x_i = y_i$ for all $i = 1, \dots, n-2$. Furthermore, $\gamma(A) = B$. If γ is even (i.e. $\gamma \in A_n$), we are done. Thus, suppose $\gamma \notin A_n$. Let $(y_{n-1} \ y_n)$ be the transposition of the elements in B . Set

$$\sigma = (y_{n-1} \ y_n)\gamma.$$

Then σ is even since $(y_{n-1} \ y_n)$ and γ are odd. Thus,

$$\begin{aligned} \sigma x_i &= (y_{n-1} \ y_n)\gamma x_i \\ &= (y_{n-1} \ y_n)y_i \\ &= y_i \quad \forall i = 1, \dots, n-2 \quad (\because y_i \notin B = \{y_{n-1}, y_n\}). \end{aligned}$$

Therefore, $\sigma x_i = y_i$ for all $i = 1, \dots, n-2$ and $\sigma \in A_n$. Hence, A_n acts $(n-2)$ -transitively. □

Abelian Groups

Exercise 0.27. (Exercise 42)

Viewing $\mathbb{Z}, \mathbb{Q}$ as additive groups, show that $\mathbb{Q}/\mathbb{Z}$ is a torsion group, which has one and only one subgroup of order n for each integer $n \geq 1$, and that this subgroup is cyclic.

Proof. We have

$$\mathbb{Q}/\mathbb{Z} = \{x + \mathbb{Z} : x \in \mathbb{Q}\}.$$

Let $x + \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ be an arbitrary nonzero element. Let $x = \frac{a}{b} \in \mathbb{Q}$. Then

$$b(x + \mathbb{Z}) = b\left(\frac{a}{b} + \mathbb{Z}\right) = a + \mathbb{Z} = 0 + \mathbb{Z} = 0.$$

Therefore, $x + \mathbb{Z}$ has finite order. Then, since $x + \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ was arbitrary, every element of $\mathbb{Q}/\mathbb{Z}$ is a torsion element and, therefore, $\mathbb{Q}/\mathbb{Z}$ is a torsion group. For each integer $n \geq 1$, define

$$H_n = \left\langle \frac{1}{n} + \mathbb{Z} \right\rangle.$$

Then $n\left(\frac{1}{n} + \mathbb{Z}\right) = 1 + \mathbb{Z} = 0$. Thus, $\frac{1}{n} + \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ has order dividing n . If $0 < k < n$, then

$$k\left(\frac{1}{n} + \mathbb{Z}\right) = \frac{k}{n} + \mathbb{Z}$$

and $\frac{k}{n} \notin \mathbb{Z}$. Hence, $\frac{1}{n} + \mathbb{Z}$ has order exactly n . Thus,

$$H_n = \left\langle \frac{1}{n} + \mathbb{Z} \right\rangle \leq \mathbb{Q}/\mathbb{Z}$$

is a cyclic subgroup of order n . Suppose $K_n \leq \mathbb{Q}/\mathbb{Z}$ is a subgroup of order n . Thus, let $x + \mathbb{Z}$ be a generator for K_n . Write $x = \frac{a}{b} \in \mathbb{Q}$ in lowest terms. Then $x + \mathbb{Z}$ has order $b = n$. Thus, every generator of a subgroup of order n has the form

$$\frac{a}{n} + \mathbb{Z}$$

with $(a, n) = 1$. Then a is invertible modulo n and thus there exists some $c \in \mathbb{Z}$ such that

$$ac \equiv 1 \pmod{n}$$

Thus $ac = 1 + nk$ with $k \in \mathbb{Z}$. Hence,

$$c \left(\frac{a}{n} + \mathbb{Z} \right) = \frac{ca}{n} + \mathbb{Z} = \frac{1 + nk}{n} + \mathbb{Z} = \frac{1}{n} + \mathbb{Z}.$$

Therefore,

$$\left\langle \frac{a}{n} + \mathbb{Z} \right\rangle = \left\langle \frac{1}{n} + \mathbb{Z} \right\rangle = H_n$$

and thus $\mathbb{Q}/\mathbb{Z}$ has one subgroup of order n , namely H_n , which is cyclic.

□